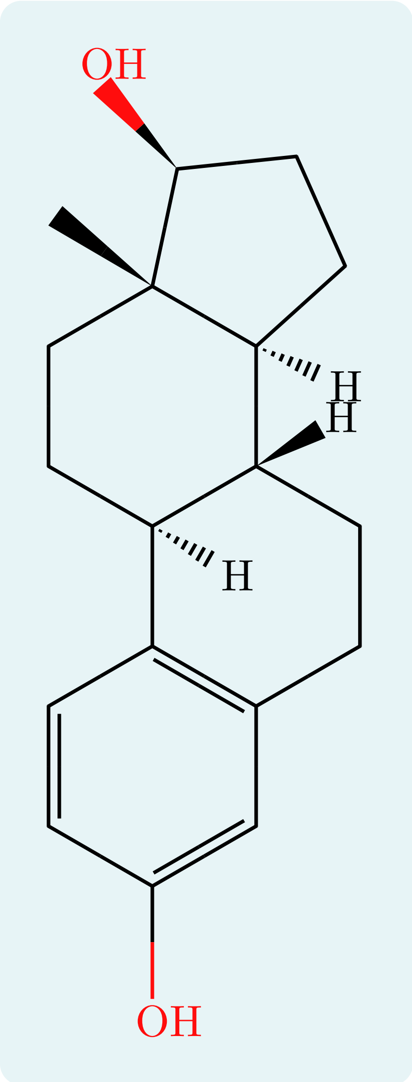
typed-smiles

Render SMILES strings as 2D molecular diagrams

User Guide



Version 0.2.0

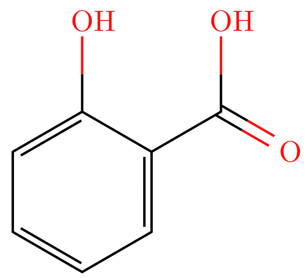
Contents

1 Introduction

typed-smiles renders SMILES strings as 2D skeletal molecular diagrams inside Typst documents. A bundled WebAssembly plugin parses the SMILES, computes atom coordinates, implicit hydrogens, and stereochemistry, and the Typst layer draws the bonds, atom labels, colors, charges, hydrogens, wedges, and reaction helpers.

This guide documents every function, argument, and package-specific syntax extension. Each feature comes with a runnable example: the source is on the left, its rendered output on the right.

Typst
```
#smiles("OC1=CC=CC=C1C(=O)O")
```



2 Getting started

2.1 Installation and import

Import the package from the Typst preview namespace. A wildcard import gives you every public symbol:

```typst
#import "@preview/typed-smiles:0.2.0": *
```

Or import only what you need:

```typst
#import "@preview/typed-smiles:0.2.0": smiles, ce, mol, rxn-arrow, reaction
```

The package exports five symbols:

Symbol	Purpose
`smiles`	Render a SMILES string as a molecule (the main function).
`ce`	Chemical formulas and equations, re-exported from `chemformula`.
`mol`	Wrap a molecule with an optional caption.
`rxn-arrow`	A reaction arrow with conditions above and below.
`reaction`	Lay out a multi-step reaction scheme.

2.2 Your first molecule

Call `smiles` with a SMILES string.

```typst
Ethanol: #smiles("CCO")
```

Ethanol:

3 The `smiles()` function

`smiles` is the main function. Its full signature is:

```typst
#smiles(
  smiles-str,
  scale: 1.0,
  bond-length: none,
  font-size: none,
  font: "New Computer Modern",
  bond-stroke: none,
  color: true,
  rotation: 0deg,
  show-all-h: false,
)
```

3.1 Argument reference

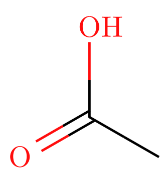
Argument	Type	Default	Description
`smiles-str`	`str`	(required)	The SMILES string.
`scale`	`float`	`1.0`	Balanced scale for bond length, label size, and stroke at once.
`bond-length`	`float` / `none`	`none`	Bond length factor (`1.0` is 30 pt). Overrides `scale` for length.
`font-size`	`length` / `none`	`none`	Atom-label font size. Overrides `scale` for labels.
`font`	`str`	`"New Computer Modern"`	Font family for atom labels.
`bond-stroke`	`length` / `none`	`none`	Bond stroke width. Overrides `scale` for strokes.
`color`	`bool`	`true`	Apply Jmol CPK atom colors.
`rotation`	`angle`	`0deg`	Rotate the molecule; labels stay upright.
`show-all-h`	`bool`	`false`	Also label carbon implicit hydrogens.

> **Note.** `scale` sizes everything together. Use `bond-length`, `font-size`, and `bond-stroke` to override one dimension on its own; each defaults to `none`, meaning it follows `scale`.

3.2 `smiles-str`

The positional argument is the SMILES string.

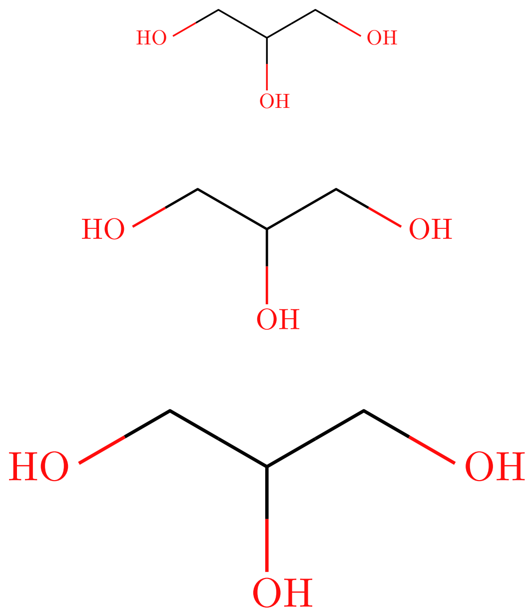
```typst
#smiles("CC(=O)O")
```



3.3 `scale`

Scales bond length, labels, and stroke proportionally.

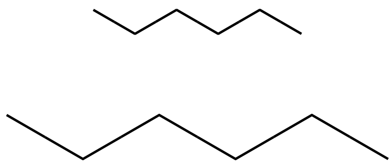
```
#smiles("OCC(O)CO", scale: 0.7) \
#smiles("OCC(O)CO", scale: 1.0) \
#smiles("OCC(O)CO", scale: 1.4)
```



3.4 `bond-length`

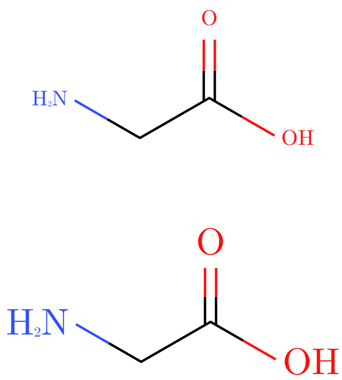
Sets bond length on its own (`1.0` is 30 pt per bond), leaving label size and stroke under `scale`.

```
#smiles("CCCCCC", bond-length: 0.6) \
#smiles("CCCCCC", bond-length: 1.1)
```



3.5 `font-size`

```
#smiles("OC(=O)CN", font-size: 8pt) \
#smiles("OC(=O)CN", font-size: 14pt)
```

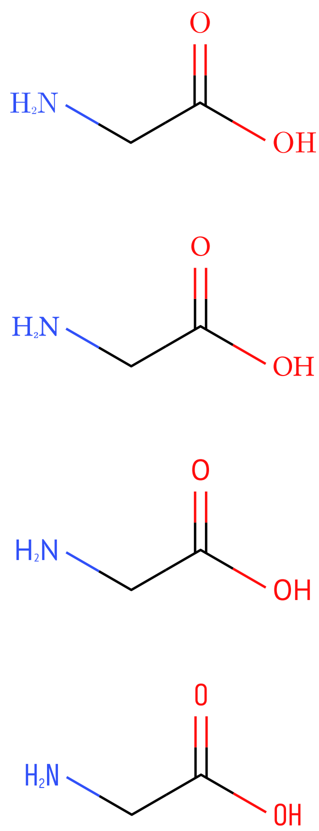


3.6 `font`

Match the document typeface, or pick something with character.

```typst
#smiles("OC(=O)CN", font: "New Computer Modern") \
#smiles("OC(=O)CN", font: "Libertinus Serif") \
#smiles("OC(=O)CN", font: "PT Sans") \
#smiles("OC(=O)CN", font: "Iosevka")
```

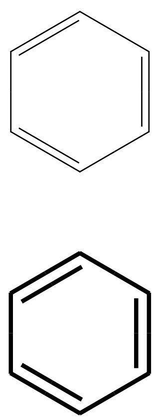
Typst



3.7 `bond-stroke`

```typst
#smiles("C1=CC=CC=C1", bond-stroke: 0.5pt) \
#smiles("C1=CC=CC=C1", bond-stroke: 1.6pt)
```

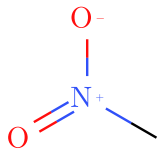
Typst

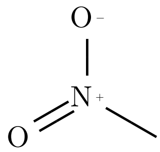


3.8 color

CPK colors are on by default (see Section 7). Set `color: false` for a monochrome diagram.

```
#smiles("C[N+](=O)[O-]") \
#smiles("C[N+](=O)[O-]", color: false)
```

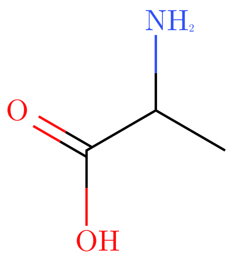


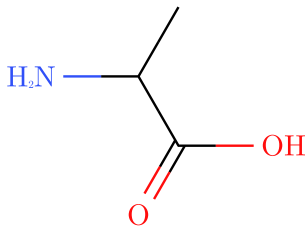


3.9 rotation

Rotates the whole molecule while keeping every label upright.

```
#smiles("CC(N)C(=O)O", rotation: 0deg) \
#smiles("CC(N)C(=O)O", rotation: 90deg)
```

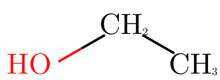




3.10 show-all-h

Hydrogens on heteroatoms are shown by default; carbon hydrogens are hidden. Set `show-all-h: true` to label every implicit hydrogen.

```
#smiles("CCO") \
#smiles("CCO", show-all-h: true)
```



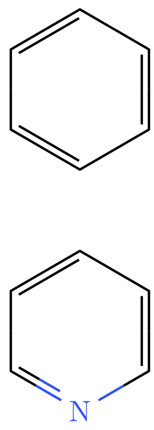
4 Atoms and charges

Standard SMILES syntax (atoms, bonds, branches, ring closures) is supported. This section covers the points specific to typed-smiles.

4.1 Aromatic rings

> **Note.** The bundled parser does not yet read lowercase aromatic atoms (`c1ccccc1`). Write aromatic rings in Kekulé form: uppercase atoms with explicit alternating double bonds.

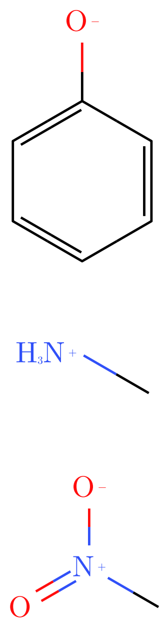
```
#smiles("C1=CC=CC=C1") \
#smiles("C1=CC=NC=C1")
```



4.2 Charges

Formal charges inside brackets render as a raised sign after the atom.

```
#smiles("[O-]C1=CC=CC=C1") \
#smiles("C[NH3+]") \
#smiles("C[N+](=O)[O-]")
```



> **Note.** A bracketed element such as `[N]` is a nitrogen atom. To print an arbitrary text label instead, use the abbreviation syntax `{N}` (see Section 8).

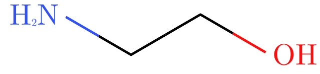
5 Hydrogens

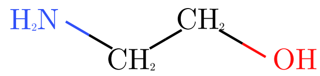
5.1 Default display

Heteroatom hydrogens are shown by default; carbon hydrogens are hidden for a clean skeleton. `show-all-h: true` labels carbon hydrogens as well.

Typst

```
#smiles("OCCN") \
#smiles("OCCN", show-all-h: true)
```



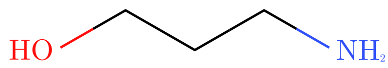


5.2 Label orientation

Terminal heteroatom labels orient so the bond meets the heavy atom rather than the trailing hydrogen, at any angle.

Typst

```
#smiles("NCCCO")
```



6 Stereochemistry

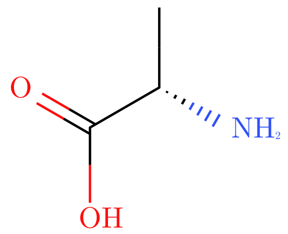
typed-smiles supports both standard SMILES stereo notations, plus a manual override for drawing wedges directly.

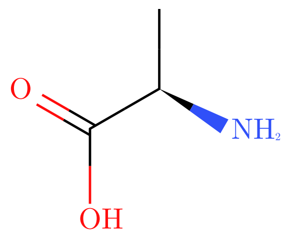
6.1 Tetrahedral centers: `@` and `@@`

A bracket atom carrying `@` or `@@` becomes a wedge (toward the viewer) or hashed bond (away), computed from the 2D layout so the depiction matches the SMILES in the conventional orientation.

Typst

```
#smiles("N[C@@H](C)C(=O)O") \
#smiles("N[C@H](C)C(=O)O")
```

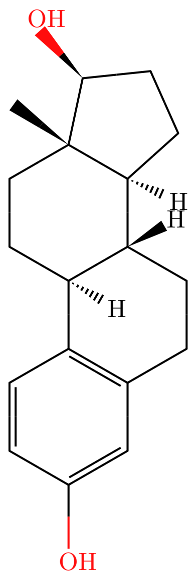




The renderer wedges an exocyclic substituent (such as OH or CH_3) where one exists, and an explicit hydrogen otherwise. Fused systems work too.

Typst

```
#smiles(
  "C[C@]12CC[C@H]3[C@H]
  ([C@@H]1CC[C@@H]2O)CCC4=C3C=CC(=C4)O",
  scale: 0.85,
)
```



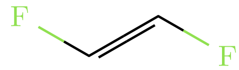
Note. The geometry is drawn correctly, but R/S descriptors are not computed. Ring stereochemistry between adjacent centers and bridged bicyclics may need a manual adjustment (see Section 11).

6.2 Double bonds: `/` and `\`

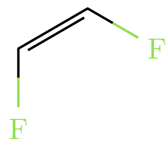
Directional bonds `/` and `\` around a double bond set its cis/trans geometry. They come in pairs, one on each side of the `=`.

```typst
trans: #smiles("F/C=C/F")
#h(2em)
cis: #smiles("F/C=C\F")
```
Typst

trans:



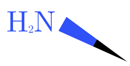
cis:

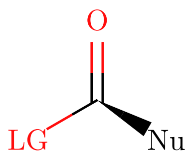


6.3 Manual wedges: `!w` and `!h`

For direct control, `!w` draws a solid wedge and `!h` a hashed wedge. These are typed-smiles extensions, not standard SMILES. Like plain bonds, they are colored by the atoms at each end.

```typst
#smiles("C!wN") \
#smiles("C!hN") \
#smiles("{Nu}!wC({LG|red})=O")
```
Typst





7 Colors

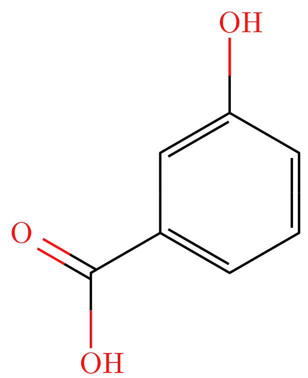
Atoms use the Jmol CPK palette. Carbon and unlisted elements are black; common heteroatoms get their conventional colors.

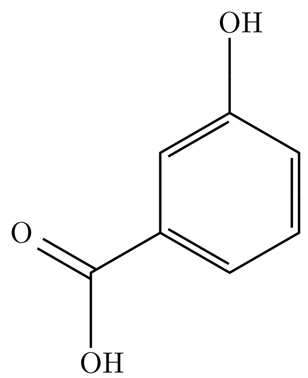
N	O	S	P	F	Cl	Br	I

Set `color: false` for a monochrome diagram.

```typst
#smiles("OC1=CC(=CC=C1)C(=O)O") \
#smiles("OC1=CC(=CC=C1)C(=O)O", color: false)
```

Typst



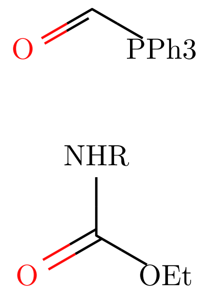


8 Abbreviated groups

8.1 Plain labels

Wrap text in braces `{...}` to place a labeled pseudo-atom that bonds like any other atom. Use it for groups you do not want to draw in full.

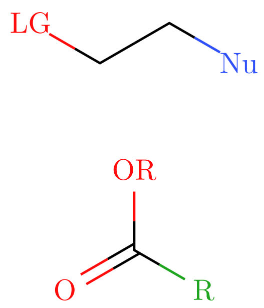
```typst
#smiles("{PPh3}C=O") \
#smiles("{OEt}C(=O){NHR}")
```



8.2 Colored labels

Add `|style` inside the braces to color a label. The styles are `red`, `blue`, `green`, `black`, `gray` (or `grey`), `orange`, and `purple`. Any element symbol also works and uses that element's color.

```typst
#smiles("{Nu|blue}CC{LG|red}") \
#smiles("{R|green}C(=O){OR|O}")
```



9 Chemical formulas: `ce()`

For formulas and text equations, `ce` is re-exported from the chemformula package, so one import covers both structures and formulas.

```typst
#ce("H2SO4") #h(1em) #ce("Ca^2+")
#h(1em) #ce("2 H2 + O2 -> 2 H2O")
```

Typst

H_2SO_4 Ca^{2+} $2\,H_2 + O_2 \longrightarrow 2\,H_2O$

9.1 Fonts and size

`ce` accepts `font` and `font-size` any other arguments pass through to chemformula.

```typst
#ce("CuSO4 * 5 H2O") \
#ce("CuSO4 * 5 H2O", font: "Libertinus
Serif", font-size: 13pt) \
#ce("CuSO4 * 5 H2O", font: "PT Sans",
font-size: 13pt)
```

Typst

$CuSO_4 \cdot 5\,H_2O$
$CuSO_4 \cdot 5\,H_2O$
$CuSO_4 \cdot 5\,H_2O$

10 Reaction schemes

Three helpers compose molecules, formulas, and arrows into schemes.

10.1 `mol()`

`mol(content, label: none)` wraps any content and centers an optional caption beneath it.

```typst
#mol(smiles("CCO"), label:
[ethanol])
```

Typst

ethanol

10.2 `rxn-arrow()`

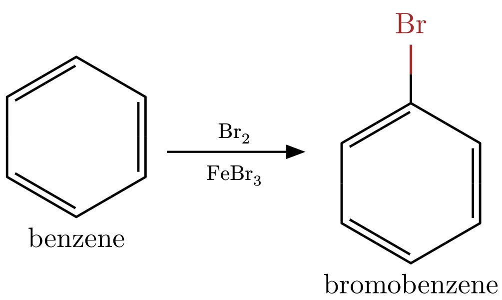
`rxn-arrow(above: none, below: none, dir: "right")` draws an arrow. `dir` may be `"right"`, `"left"`, `"up"`, or `"down"`. `above` and `below` carry reagents and conditions.

10.3 `reaction()`

`reaction(gap-h: 1.5em, gap-v: 1.5em, ..items)` lays out molecules and arrows left to right. An up or down arrow wraps the scheme onto a new row.

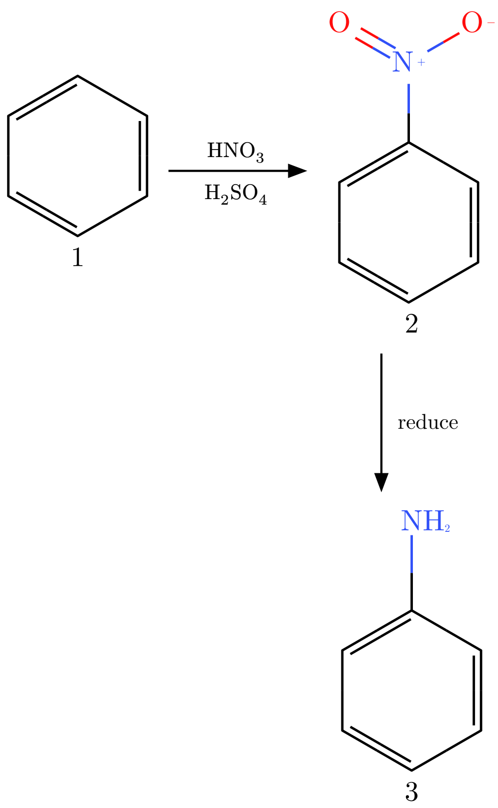
```typst
#reaction(
  mol(smiles("C1=CC=CC=C1"), label: [benzene]),
  rxn-arrow(above: ce("Br2"), below: ce("FeBr3")),
  mol(smiles("BrC1=CC=CC=C1"), label: [bromobenzene]),
)
```

Typst



A wrap-around scheme using a downward arrow:

```typst
#reaction(
  mol(smiles("C1=CC=CC=C1"), label: [1]),
  rxn-arrow(above: ce("HNO3"), below: ce("H2SO4")),
  mol(smiles("[O-][N+](=O)C1=CC=CC=C1"), label: [2]),
  rxn-arrow(dir: "down", above: [reduce]),
  mol(smiles("NC1=CC=CC=C1"), label: [3]),
)
```



11 Limitations

- Lowercase aromatic SMILES (`c1ccccc1`) are not parsed. Use Kekulé notation.
- Directional bonds (`/`, `\`) draw the correct cis/trans geometry, but E/Z descriptors are not computed.
- R/S and E/Z descriptors are not calculated.
- Ring stereochemistry between adjacent centers and bridged bicyclics can overlap or need a manual adjustment (try `rotation`, or the manual `!w` and `!h` wedges).

12 Quick reference

12.1 Syntax

Syntax	Meaning
`[...]`	Bracket atom (any element, charges, explicit H).
`@` / `@@`	Tetrahedral anticlockwise / clockwise.
`/` `\`	Double-bond cis/trans geometry.
`!w` / `!h`	Manual solid / hashed wedge (typed-smiles extension).
`{label}`	Abbreviated group pseudo-atom.
`{label|style}`	Colored abbreviated group.

12.2 `smiles()` options

Option	Default	Effect
`scale`	`1.0`	Balanced size of everything.
`bond-length`	`none`	Bond length only (`1.0` is 30 pt).
`font-size`	`none`	Atom-label size only.
`font`	`"New Computer Modern"`	Atom-label font.
`bond-stroke`	`none`	Bond width only.
`color`	`true`	CPK colors on or off.
`rotation`	`0deg`	Rotate, labels stay upright.
`show-all-h`	`false`	Label carbon hydrogens too.